B.TECH PROJECT · VIVA & UNDERSTANDING GUIDE

Metal Hydrides, Machine Learning and Application Screening

What the project really does, how each part works, why each choice was made, and how to answer when the professor cross-questions you.

Understand mode

Each component explained from zero: the physics, the data, the model and the screening, with your own project's numbers and worked examples.

Defend mode

The questions an examiner is likely to ask about each component, with short model answers, plus the traps where a careless answer would overclaim.

0 The project in five minutes

1 Hydrogen storage and metal hydrides

2 Thermodynamics: Van't Hoff

3 The data and the audit

4 Turning a formula into numbers

5 The machine-learning model

6 Screening and ranking

7 The software platform

8 Limitations and future work

9 Viva question bank

10 One-page cheat sheet

Every number comes from the project's own result files, the same ones the report uses. Tags: **Report** = a result of your project (chapter of the report); **background** = standard textbook knowledge you should know but did not produce. Report: *Machine Learning-Assisted Discovery and Application-Specific Screening of Metal Hydride Materials for Hydrogen Storage*, Avinash (230151005), supervisor Dr. Saket Verma.

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Part 0 The project in five minutes

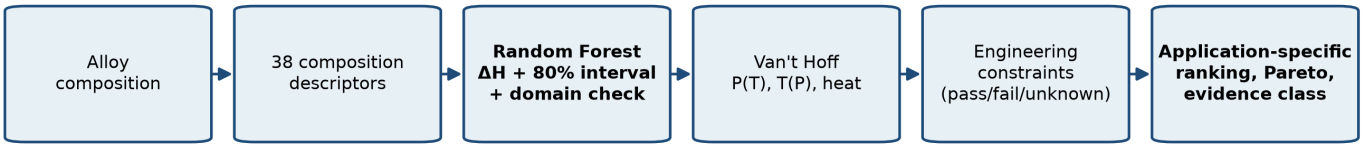
Read this part first and read it again on the morning of the viva. If you can explain this page in your own words, you can answer most questions, because almost every cross-question is about one of the arrows in the flow below.

UNDERSTAND

0.1 The story in one paragraph

Hydrogen is hard to store. Some metal alloys soak it up like a sponge and release it when heated. **Which alloy suits a given job** (a fuel-cell vehicle, a factory buffer) depends mainly on one number, the enthalpy **ΔH**, because ΔH fixes the temperature at which the alloy releases hydrogen at a useful pressure. Measuring ΔH for every possible alloy is slow, so this project (1) cleaned and audited two public databases of measured hydrides, (2) trained a machine-learning model that **estimates ΔH from the chemical formula alone** and says how much to trust each estimate, (3) used the Van't Hoff equation to turn ΔH and ΔS into **operating temperature, pressure and heat demand**, and (4) built a **screening tool** that ranks real, measured alloys against an application's requirements. Everything is packaged in an interactive app. No lab experiments were done; a lab plan is proposed for the future.

0.2 The flow of the project



Experimental ΔH, ΔS (Purdue .mh) enter at the Van't Hoff step for known materials; ML ΔH with an assumed ΔS is used only for new compositions and is labelled an estimate.

The pipeline (report Fig. 1.1). Measured ΔH and ΔS are always used when they exist; the ML estimate is only for new compositions.

Step 1 · Data: two public databases (Purdue toolbox, 337 materials; ML-HydPARK, 772 rows) → audit → one clean table of **725 compositions** with measured ΔH. Report Ch 3

↓

Step 2 · Descriptors: each formula (e.g. $LaNi_5$) → **38 numbers** computed from the elements' properties (mass, size, electronegativity …). Report §3.4

↓

Step 3 · Model: Random Forest learns descriptors → ΔH. Error **MAE 5.26 kJ/mol**. Every prediction gets an 80% interval and an "in/out of domain" flag. Report Ch 4

↓

Step 4 · Thermodynamics: Van't Hoff turns ΔH, ΔS into "at what temperature do I get 1 bar?" and "how much heat per kg H_2?". Report Ch 5

↓

Step 5 · Screening: requirements (temperature, pressure, mass …) → pass/fail/unknown → weighted score → Pareto front → ranked list with evidence class. Report Ch 6-7

↓

Step 6 · Platform: everything above inside an 8-page Streamlit app, "METAL HYDRIDE AI". Report §6.3

0.3 Twelve numbers to remember

725 compositions in the ML dataset (276 Purdue, 449 HydPARK values)	**248 / 288** Purdue compositions also in HydPARK → no external validation possible	**38** composition descriptors (inputs to the model)
5.26 ± 0.14 kJ/mol: Random Forest MAE, nested grouped CV (R^2 0.72)	**8.88** kJ/mol: MAE of the simple "family median" baseline (41% worse)	**≈ 2.2** kJ/mol: scatter between independent measurements (the floor)
2.6 → 16.9 kJ/mol: MAE near vs far from training data	**0.15** L1 distance: applicability-domain limit	**32.15 / 112.8** $LaNi_5$ ΔH (kJ/mol) / ΔS (J/mol·K) → 1 bar at 11.6 °C
85% of single-plateau materials reach 1 bar at ≤ 80 °C	**5.5 wt%** DOE 2025 *system* target; intermetallics give ≈ 1–2 wt% (material)	**77** automated software tests, all passing

0.4 What you can claim, and what you must not

YOU CAN CLAIM

- A composition-only model predicts ΔH of AB_5/AB_2-type alloys with MAE ≈ 5.3 kJ/mol, 41% better than a family-median rule, under honest (nested, grouped) validation.
- The error is largest far from the training data, and you quantified this (calibration table, domain limit).
- The public datasets have serious traps (sign conventions, back-calculated ΔS, overlap), and you found and handled them.
- For the stated requirements, Ti-based AB_2/AB alloys rank highest for mobile use, and TiFe-type/Ti–Zr–Mn alloys for stationary use.
- The workflow runs as a tested, interactive tool.

YOU MUST NOT CLAIM

- "The model discovers new materials." It only interpolates near known alloys; it fails on unseen families.
- "The model predicts operating temperature." It predicts ΔH only; ΔS is *assumed* for new alloys.
- "The 80% interval guarantees the true value is inside."
- "This alloy is the best." Rankings are screening results under stated weights.
- "2 wt% storage" as a system figure. It is material-level; a tank would be lower.
- Any experimental result. No experiments were done.

SAY THIS WHEN ASKED "EXPLAIN YOUR PROJECT IN TWO MINUTES"

"Metal hydrides store hydrogen safely at low pressure, but which alloy suits an application depends on its enthalpy ΔH, which sets the release temperature. I combined two public hydride databases, found that they overlap heavily and use opposite sign conventions, and built a clean set of 725 compositions. I trained a random forest on 38 composition-based descriptors; under grouped, nested cross-validation it predicts ΔH with a mean error of about 5 kJ/mol, which is 41% better than a simple family rule. I measured how the error grows away from the training data, so each prediction has an interval and a reliability flag. Then I used the Van't Hoff equation to get operating temperatures and heat demand, and built a screening tool with hard constraints, weighted scores and Pareto fronts that ranks measured alloys for mobile and stationary storage. It's all in an interactive app. The model is a screening aid for alloys near known families; any candidate still needs a PCT measurement."

Part 1 Hydrogen storage and metal hydrides

Report: Chapter 1 and §2.1, §2.3–2.4

The physics you need before anything else: why hydrogen is hard to store, what actually happens inside a metal hydride, and why the pressure stays flat (the "plateau") while the metal fills up. Examiners from an energy background usually start here.

UNDERSTAND

1.1 Why storing hydrogen is hard background

Hydrogen carries a lot of energy per *kilogram*: its lower heating value (LHV) is about **120 MJ/kg** (the project uses 119.96 MJ/kg), roughly three times that of petrol. But as a gas at room conditions it is extremely light, so it carries very little energy per *litre*. Every storage method is a way of packing more hydrogen into a given volume:

Method	How	Main cost
Compressed gas	squeeze to several hundred bar in a strong tank	compression energy, heavy/expensive vessel, high-pressure safety
Liquid hydrogen	cool to about 20 K (−253 °C)	liquefaction energy, boil-off, cryogenic insulation
Solid-state (metal hydride)	let a metal alloy absorb hydrogen into its crystal	weight of the metal, heat management

Metal hydrides store hydrogen at a **few bar**, close to room temperature, and can pack more hydrogen per litre than liquid hydrogen. Their weakness is **mass**: the metal is heavy, so only about 1–2% of the weight is hydrogen for the common alloys.

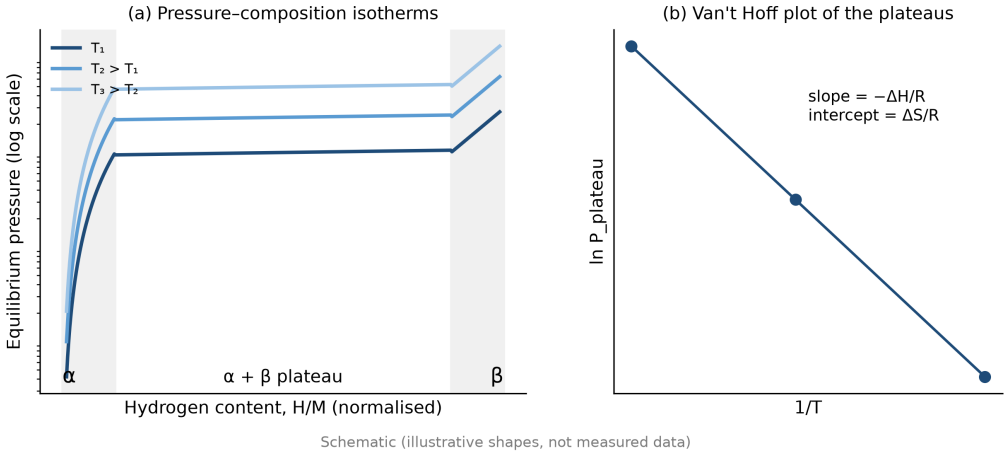
1.2 What happens inside a metal hydride background

> **PICTURE IT**
> Imagine a crystal of metal atoms as a stack of marbles. Between the marbles there are small gaps ("interstitial sites"). A hydrogen molecule H_2 lands on the surface, splits into two H atoms, and the tiny H atoms slip into the gaps. Fill enough gaps and the metal becomes a new compound, the **hydride**. Heat it and the H atoms come back out, recombine into H_2 and leave as gas.

The steps, in order: (1) H_2 sticks to the surface (physisorption); (2) it dissociates into 2 H atoms; (3) the atoms diffuse into interstitial sites. Two directions:

- **Absorption** (charging): metal + H_2 → hydride. It **releases heat** (exothermic), so the tank must be *cooled* while refuelling.
- **Desorption** (discharging): hydride → metal + H_2. It **needs heat** (endothermic), so the tank must be *heated* to deliver hydrogen. If the heat stops, the hydrogen flow stops, which is a built-in safety feature.

1.3 The α phase, the β phase and the plateau background



Schematic PCT isotherms (report Fig. 2.1). Illustrative shapes only.

Read the left plot from left to right, at one fixed temperature, as you push hydrogen in:

1. **α region** (left): a little hydrogen dissolves in the metal as a solid solution. Pressure rises quickly with each bit added.
2. **α + β plateau** (middle): the hydride phase β starts to form. Now two solid phases coexist, and the pressure stays almost **constant** while the metal converts from α to β. This flat part is where nearly all the useful storage happens.
3. **β region** (right): all metal is hydride; adding more needs a steep pressure rise.

WHY IS THE PLATEAU FLAT? (GIBBS PHASE RULE)

Phase rule: degrees of freedom $F = C - P + 2$. On the plateau there are $C = 2$ components (metal, hydrogen) and $P = 3$ phases (α solid, β solid, H_2 gas). So $F = 2 - 3 + 2 = 1$. Choosing the temperature uses up that one freedom, so the **pressure is fixed**: it cannot change while α and β coexist. That fixed pressure is the **plateau pressure** P_{eq}. At a higher temperature the plateau sits higher (the three curves in the figure).

This curve is called the **PCI** (pressure–composition isotherm) or **PCT** (pressure–composition–temperature) curve. Measuring it in a **Sieverts apparatus** (a calibrated gas volume connected to the sample) is the standard experiment.

1.4 How real materials differ from the ideal background Report §2.1

Effect	What it is	Why an engineer cares
Sloped plateau	the "flat" part tilts upward, because of composition gradients and lattice strain	the last hydrogen comes out at a lower pressure than the first. In your data the pressure at 10% charge is a median of 1.75× lower than at 90%.
Hysteresis	absorption happens at a higher pressure than desorption at the same T	you must charge at a higher pressure (compressor work). For TiFe the charging pressure is ≈ 1.9× the delivery pressure.
Several plateaus	two or more reaction steps, each with its own ΔH, ΔS (e.g. $NaAlH_4$)	only part of the capacity is available at the lower temperature
Activation	fresh alloy needs initial cycles/heating before it absorbs (TiFe is famous for this)	extra processing step

1.5 The families of hydride alloys Report Table 2.1

Alloys are named by stoichiometry. **A** = an element that likes hydrogen strongly (La, rare earths, Ti, Zr, Mg, Ca). **B** = an element that does not (Ni, Fe, Mn, Cr, Co, Al). Mixing A and B tunes how strongly the alloy holds hydrogen.

Family	Examples	Median ΔH kJ/mol	Median wt% (max)	Median T(1 bar)	Remember
AB_5	$LaNi_5$, $MmNi_{4.5}Al_{0.5}$, $CaNi_5$	33.0	1.2 (1.9)	24 °C	the reference hydride; easy activation, flat plateau; La/Co costly
AB_2 (Laves)	$TiCr_{2-x}Mn_x$, $ZrMn_2$, Ti–Zr–Mn–V	29.0	1.7 (2.9)	−4 °C	largest family; lighter, so higher wt%; very tunable; sloped plateaus
AB	TiFe, $TiFe_{1-x}Mn_x$	34.7	1.5 (2.2)	32 °C	cheap elements; hard to activate
SS (BCC)	V–Ti–Fe, V–Ti–Cr	30.8	0.7 (3.7)	49 °C	up to ≈ 4 wt%, but only part is reversible at ambient
Mg-based	MgH_2, Mg_2Ni	72.4	5.6 (7.6)	266 °C	high capacity, but needs ≈ 300 °C and is slow
Complex	$NaAlH_4$ (+Ti)	two steps: 37 and 47 kJ/mol			reversible only with Ti doping; two plateaus

Numbers are medians over single-plateau Purdue materials, computed in this project from the measured ΔH, ΔS and reported capacities.

THE ONE TRADE-OFF THAT EXPLAINS EVERYTHING

Light alloys (Mg-based) store more hydrogen per kg but hold it too tightly (high ΔH), so they need high temperatures. Alloys that release hydrogen near room temperature (AB_5, AB_2, AB) are transition-metal alloys: heavy, so only ≈ 1–2 wt%. **No material in the data has both high capacity and near-ambient operation.**

1.6 Engineering properties and their definitions Report §2.3

- **Gravimetric capacity**, wt% = mass of H ÷ mass of hydride × 100. Also given as **H/M**: hydrogen atoms per metal atom, or per formula unit (the Purdue files use per formula unit). Literature values differ in definition: total uptake, reversible capacity between two pressures, or first plateau only.
- **Material vs system capacity**: a real tank adds the vessel, a heat exchanger and empty space between powder grains, so the *system* capacity is much lower than the material's. The **DOE 2025 target of 5.5 wt% is a system target**.
- **Volumetric capacity** (kg H_2 per m^3) needs the density of the hydride or the powder bed. It is reported for only 37 Purdue materials, so the project does not estimate it elsewhere.
- **Reaction heat** Q (MJ per kg H_2): must be supplied on discharge and removed on refuelling.
- **Kinetics**: how fast it absorbs/desorbs. Depends on particle size, surface, catalysts, and how far the pressure is from equilibrium.
- **Cycle life**: capacity loss over many cycles. Not available in the datasets.

DEFEND

Q1.1 Why metal hydrides and not compressed gas?

Hydrides store at a few bar near room temperature, so the vessel is cheaper and safer, and the volumetric density is high. The price is weight (≈ 1–2 wt% for common alloys) and the need to manage heat. So they suit stationary storage, forklifts or systems with waste heat, where weight matters less.

Q1.2 What is the plateau and why is it flat?

It is the two-phase region where α (solid solution) and β (hydride) coexist. With 2 components and 3 phases, the phase rule gives one degree of freedom, so at a fixed temperature the pressure is fixed. That is the plateau pressure, P_{eq}.

Q1.3 What is hysteresis and why does it matter?

Absorption needs a higher pressure than desorption at the same temperature. Each cycle therefore needs extra compression (or a larger temperature swing), which is an energy loss. For TiFe the charging pressure is about 1.9 times the desorption pressure.

Q1.4 Is 2 wt% good enough for a car?

Not for a light-duty car. The DOE 2025 target is 5.5 wt% for the whole *system*, and ≈ 2 wt% is the *material* alone; a system would be lower still. That is why the report recommends hydrides for uses where mass matters less: forklifts, buses, stationary buffers.

TRAP Never present 2 wt% as a tank figure.

Q1.5 Why not just use MgH_2? It holds 7.6 wt%.

Its ΔH is ≈ 78 kJ/mol, so it needs ≈ 300 °C to give 1 bar, and the heat needed is ≈ 39 MJ per kg H_2, almost a third of the hydrogen's own energy content. It is also kinetically slow. Fuel-cell waste heat (≈ 70 °C) cannot drive it, so Mg-based materials fail both screens.

Q1.6 Why is absorption exothermic?

Hydrogen atoms bound in the lattice are at lower energy than free H_2 gas, so forming the hydride releases energy. Getting them out costs the same energy, which is why desorption needs heat. The amount per mole of H_2 is ΔH.

Part 2 Thermodynamics: the Van't Hoff equation

Report: §2.2 and Chapter 5

One equation connects the two numbers ΔH and ΔS to everything an engineer wants to know: the pressure at a given temperature, the temperature needed for a given pressure, and the heat to supply. Know how to derive it, how to use it with real numbers, and what its limits are.

UNDERSTAND

2.1 What ΔH and ΔS mean physically

ΔH: ENTHALPY OF DECOMPOSITION

The energy needed to pull one mole of H_2 out of the hydride (kJ per mol H_2). A **bigger ΔH** means hydrogen is held **more tightly**, so a higher temperature is needed to release it. $LaNi_5$: 32 kJ/mol; MgH_2: 78 kJ/mol.

ΔS: ENTROPY OF DECOMPOSITION

Mostly the entropy hydrogen gains when it goes from "locked in the crystal" to "free gas". The standard entropy of H_2 gas is ≈ 130.7 J/(mol·K), so ΔS is similar for most hydrides. In your data the median is **110.6 J/(mol·K)**, with half of all materials between 102.8 and 118.9.

SIGN CONVENTION TRAP

Chemists often quote **formation** values (metal + H_2 → hydride), which are **negative** (−32 kJ/mol). This project uses **positive desorption** values (+32 kJ/mol). Same physics, opposite sign. The Purdue files store negative values in J/mol; HydPARK stores positive values in kJ/mol. Mixing them unconverted would be a disaster, so the audit converted everything to positive kJ/mol (ΔH) and J/(mol·K) (ΔS).

2.2 Deriving the Van't Hoff relation

For desorption, $\frac{2}{x}MH_x \rightleftharpoons \frac{2}{x}M + H_2$. The Gibbs energy change, with hydrogen at pressure P, is

$$\Delta G = \Delta H - T\Delta S + RT \ln \frac{P}{P_0}.$$

At equilibrium on the plateau $\Delta G = 0$, and $P = P_{eq}$:

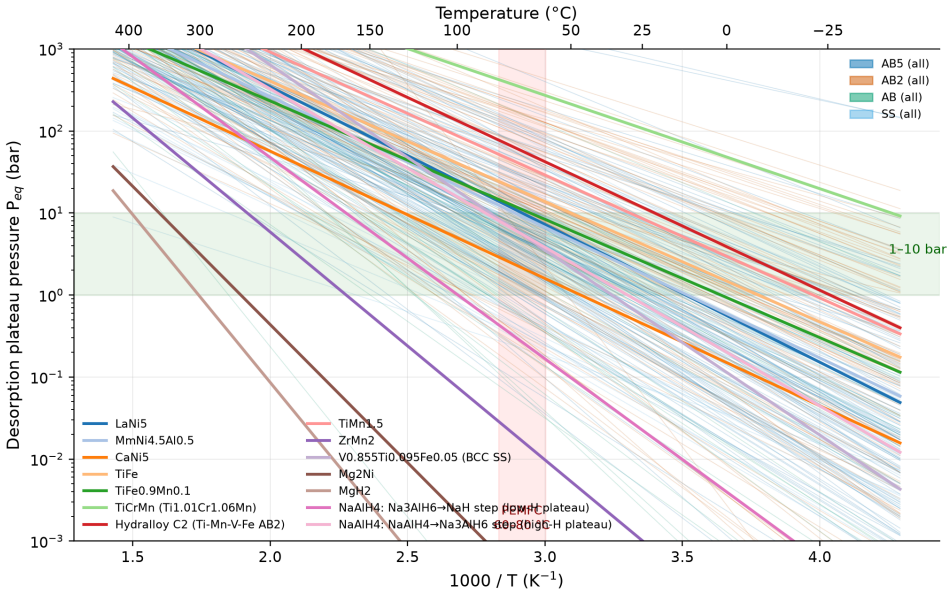
$$\boxed{\ln \frac{P_{eq}}{P_0} = -\frac{\Delta H}{RT} + \frac{\Delta S}{R}}$$

$R = 8.314462618$ J/(mol·K), and $P_0 = 1$ atm $= 1.01325$ bar is the reference pressure (both datasets use 1 atm). Two rearrangements are used constantly:

$$T(P) = \frac{\Delta H}{\Delta S - R\ln(P/P_0)}, \qquad Q = \frac{\Delta H}{M_{H_2}},\ M_{H_2} = 2.01588\ \text{g/mol}.$$

HOW TO READ IT

Plot $\ln P_{eq}$ against $1/T$: a straight line, slope $-\Delta H/R$, intercept $\Delta S/R$. That is why measuring PCT curves at **several temperatures** gives ΔH and ΔS: take each curve's plateau pressure, plot, fit a line.



Van't Hoff lines of representative materials (report Fig. 5.1). Steeper line = larger ΔH. The green band is 1–10 bar.

2.3 Worked example: $LaNi_5$ Report §5.1

$LANI_5$, ΔH = 32.15 KJ/MOL, ΔS = 112.8 J/(MOL·K)

Pressure at 25 °C (298.15 K):

$$\ln \frac{P}{\text{atm}} = -\frac{32151}{8.314 \times 298.15} + \frac{112.8}{8.314} = -12.970 + 13.567 = 0.597$$

so $P = e^{0.597} = 1.817$ atm $= 1.84$ bar. **$LaNi_5$ gives ≈ 1.8 bar at room temperature.**

Temperature for 1 bar: $\ln(1/1.01325) = -0.0132$, so

$$T = \frac{32151}{112.8 + 8.314 \times 0.0132} = \frac{32151}{112.91} = 284.8\ \text{K} = 11.6\ ^\circ\text{C}.$$

Heat per kg H_2: $Q = 32.15/2.01588 = 15.95$ MJ/kg, which is $15.95/119.96 = 13.3\%$ of the LHV. **To release 1 kg of H_2 you must supply ≈ 16 MJ of heat, and about the same is released when refilling.**

2.4 Three rules of thumb

1. $T(1\text{ bar}) \approx \Delta H/\Delta S$. LaNi5: 32150/112.8 = 285 K ✓. Since ΔS is roughly constant, **ΔH sets the operating temperature.**
2. **5 kJ/mol in ΔH ≈ a factor of 7.5 in pressure** at 25 °C: $e^{5000/(8.314\times298)} = e^{2.02} \approx 7.5$. This is why a model error of 5 kJ/mol is fine for sorting alloys into classes but too coarse to design a plateau pressure.
3. **Heat demand ∝ ΔH.** 1 kJ/mol ≈ 0.496 MJ/kg H_2. Near-ambient intermetallics need ≈ 14–16 MJ/kg (11–13% of the LHV); MgH_2 needs 38.8 MJ/kg (32.4%).

2.5 What the analysis found across the database Report §5.2-5.3

Material	ΔH	P(25 °C) bar	T(1 bar) °C	Q MJ/kg	Meaning
$LaNi_5$	32.2	1.84	11.6	15.9	room-temperature reference
TiFe	28.1	4.16	−8.3	13.9	cheap, low-pressure
TiCrMn	21.8	107.5	−78.5	10.8	high-pressure AB_2, needs strong vessel
MgH_2	78.3	2.5×10^{-7}	302	38.8	far too stable for ambient use
Mg_2Ni	64.5	1.2×10^{-5}	255	32.0	same problem

- Of 318 single-plateau materials with typical ΔS, **85% reach 1 bar at ≤ 80 °C** (PEM fuel-cell waste heat is enough), 53% at ≤ 25 °C, and only 2% need more than 150 °C.
- Materials that work with ≤ 80 °C heat need a median of **15.1 MJ/kg**; those needing more than 80 °C need 22.4 MJ/kg. **Temperature and heat demand go together** (report Fig. 5.2).

2.6 Hysteresis, slope and the $NaAlH_4$ check

Report §5.4-5.5

Hysteresis: 82 materials state absorption ΔH, ΔS and 41 more state a hysteresis constant; in every case $P_{abs} \geq P_{des}$ (no inconsistent record). Median ratio at 25 °C: 1.73.

NaAlH4 (two plateaus): the Purdue file gives two steps, 37 and 47 kJ/mol. Which plateau is which? The digitised isotherm at 433 K settles it: the low-hydrogen plateau (6.1 bar) matches the 47 kJ/mol step Na3AlH6 → NaH (predicted 8.27 bar, ratio 0.74); the high-hydrogen plateau (77.8 bar) matches the 37 kJ/mol step NaAlH4 → Na3AlH6 (predicted 82.2 bar, ratio 0.95). The opposite assignment would be off by ≈ 10×. Consequence: only the first step works near 30 °C; using the full 3.9 wt% needs ≈ 100 °C.

PCI consistency: refitting Van't Hoff lines to the digitised isotherms reproduces the tabulated ΔH within a median of **1.76 kJ/mol** (53% within 2 kJ/mol). This is a *consistency check*, not independent validation, because for 32 of the fits the toolbox authors derived the tabulated value from those same curves.

2.7 Estimates for new alloys: ML ΔH + assumed ΔS

Report §5.6

For an alloy with no measurement, ΔH comes from the model (with its 80% interval) and ΔS is **assumed** to be the median of its family (e.g. AB_5: 111.2). The two uncertainties combine into a wide temperature band: about **105 K wide**, ≈ 65 K from ΔH and ≈ 35 K from ΔS. Example: $LaNi_{4.2}Al_{0.8}$, predicted ΔH 38.7 kJ/mol (34.3–41.6), gives T(1 bar) ≈ 74 °C with a band of 15–124 °C. So an ML estimate can say "more stable than $LaNi_5$"; it cannot fix an operating window.

DEFEND

Q2.1 Derive the Van't Hoff equation.

$\Delta G = \Delta H - T\Delta S + RT\ln(P/P_0)$; at equilibrium $\Delta G = 0$, so $\ln(P_{eq}/P_0) = -\Delta H/RT + \Delta S/R$. (Write it on the board; it takes 30 seconds.)

Q2.2 What assumptions does it make?

(1) ΔH and ΔS do not change with temperature over the range used; (2) a flat plateau (single equilibrium pressure); (3) ideal-gas behaviour of H_2 at the pressures used. For strongly sloped plateaus, (2) fails, and a window computed from the mid-plateau is optimistic at low state of charge.

Q2.3 Why is ΔS roughly constant across hydrides?

It is dominated by the loss of the gas-phase entropy of H_2 (≈ 130.7 J/mol·K standard). The solid-state contributions differ between alloys but are smaller. In the Purdue data the interquartile range is 102.8–118.9 J/mol·K.

Q2.4 Why do you use P_0 = 1 atm and not 1 bar?

Because both datasets define ΔS with respect to 1 atm. The toolbox's MATLAB function uses 101 325 Pa, and HydPARK's ln P(25 °C) column is consistent with 1 atm. Using 1 bar would shift every pressure by 1.3%. Results are then converted to bar.

Q2.5 Your $NaAlH_4$ table has two rows. Why?

NaAlH4 decomposes in two steps with different ΔH (37 and 47 kJ/mol). I checked the assignment against the digitised 433 K isotherm: ratios of 0.74 and 0.95 between curve and prediction; the reverse assignment would be ≈ 10× off.

Q2.6 Why is heat demand important?

Desorbing 1 kg H_2 from $LaNi_5$ needs ≈ 16 MJ, and refuelling releases about the same, which must be removed quickly. Powder beds conduct heat poorly, so heat transfer often limits the flow rate. For MgH_2 the heat is ≈ 32% of the hydrogen's LHV, a large efficiency penalty.

Q2.7 How accurate is the T(1 bar) you report for a new alloy?

Not very: the band is ≈ 105 K wide, because the ΔH interval and the assumed ΔS both matter. That is exactly why new alloys are presented as candidates that need a PCT measurement.

TRAP Don't present an estimated T(1 bar) as a property of the alloy.

Part 3 The data and the audit

Report: Chapter 3 (§3.1–3.3), Appendix C

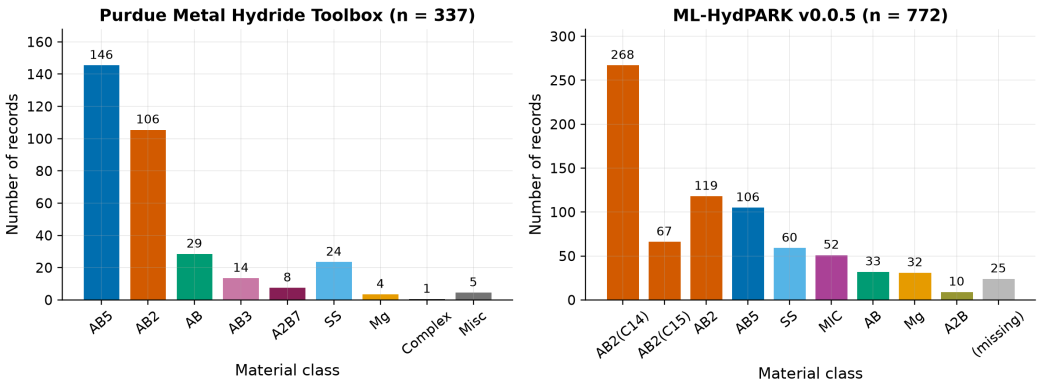
The most defensible part of your project, and the part most students skip. Know what was in each dataset, what was wrong with it, and how each problem would have produced a misleading model if left unfixed.

UNDERSTAND

3.1 The two datasets

	Purdue Metal Hydride Toolbox	ML-HydPARK v0.0.5
Who	Purdue Hydrogen Systems Lab (Voskuilen, Waters, Pourpoint 2014), MATLAB toolbox on GitHub	Sandia National Laboratories; cleaned version of the old HydPARK database for ML (Witman et al. 2020, Zenodo 2024)
Records	337 material files (.mh)	772 rows, in 4 appended blocks (397 / 25 / 15 / 335)
Content	ΔH, ΔS per plateau (absorption too for 83), capacity, slope, hysteresis, some heat-transfer and kinetics data, reference; 68 digitised PCT curves	class, formula, wt%, ΔH, ΔS, one (T, P) point, derived P(25 °C), H/M, DOI
Sign / unit	negative formation values, J/mol H_2	positive values, kJ/mol H_2
Largest families	AB_5 146, AB_2 106, AB 29, SS 24	AB_2 (incl. C14/C15) 454, AB_5 106, SS 60, MIC 52

Both are **compilations of published measurements**. Neither contains new experiments. Raw files were copied into `data/raw/` and never modified; every change happens in derived copies.



Material classes in the two datasets (report Fig. 3.1).

3.2 What the audit found, and why each finding matters Report §3.2

Finding	What it means	If ignored …
HydPARK has 772 rows, not ≈ 420	the file is four blocks appended one after another; the index restarts three times	wrong dataset size and provenance
Opposite signs and units	Purdue −J/mol, HydPARK +kJ/mol	merged ΔH values would be nonsense (−32151 vs +32.15)
HydPARK ΔS is back-calculated	for 395 of the 412 rows with a (T, P) point, $\Delta S = R \ln P + \Delta H/T$ holds to 0.01 J/mol·K: ΔS was *computed* from ΔH	**target leakage**: a model using ΔS as input would look almost perfect but be useless (see box)
"H/M" means per formula unit	found by reading the MATLAB loader `ReadThermo.m`	capacities off by the number of atoms in the formula
Capacity definition unclear	only 8 files say "reversible"; 17 say assumed / approximate / theoretical	capacity treated as more precise than it is

Toolbox fills gaps with defaults	missing hysteresis, density, kinetics are silently replaced by family "type" values	invented numbers treated as measurements. Defaults were excluded; missing stays missing.
All 68 PCT files digitised	traced from published figures; 14 shifted by the curators	treating them as raw data
Reference / formula problems	102 files cite an unverified Sandia placeholder; 49 use pseudo-elements (mischmetal Mm, Lm, ferrovanadium Vf); 12 are multi-plateau	unparseable formulas, unknown provenance
Heavy overlap	**248 of 288** parseable Purdue compositions are also in HydPARK, mostly with the same value (median difference 0.0)	"external validation" would just test memorisation

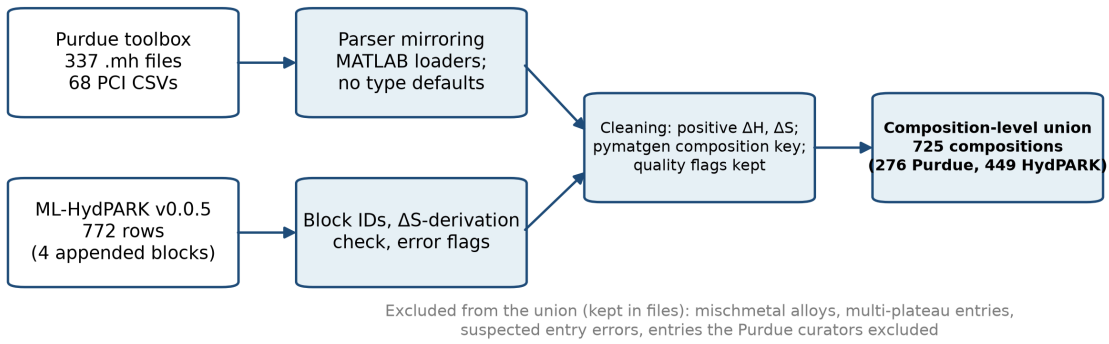
TARGET LEAKAGE, EXPLAINED SIMPLY

Suppose you want to predict ΔH, and one of your inputs is ΔS. In HydPARK, ΔS was calculated as $R \ln P + \Delta H/T$ from ΔH and one (T, P) point. So if the model knows ΔS, T and P, it can recover ΔH exactly. It would score a near-perfect error, but for a *new* alloy you never have ΔS in advance, so the model would be useless. The fix: **no thermodynamic, pressure, temperature or capacity value is allowed as an input**. Only the chemical formula. The feature-building script checks this automatically.

WHY THE OVERLAP KILLS "EXTERNAL VALIDATION"

A common idea is: train on Purdue, test on HydPARK. But HydPARK copied most Purdue values from the same papers. Testing on them is like setting an exam with the questions the student already studied: a high score proves memory, not understanding. So the project uses **one combined dataset with grouped cross-validation** instead (Part 5). It also uses the overlap usefully: where two sources report the *same composition independently*, they differ by ≈ **2.2 kJ/mol**. That is the measurement scatter, the best error any model can hope for.

3.3 Building the modelling dataset Report §3.3



From two raw sources to one composition-level table (report Fig. 3.2).

1. Parse every formula with **pymatgen** into atomic fractions (LaNi5 → La 0.167, Ni 0.833). The rounded fractions form a **composition key**, so "LaNi5" and "La1Ni5" are recognised as the same alloy.
2. Keep **one row per composition**. If Purdue has a single-plateau value, use it; otherwise use the median of the HydPARK values.
3. Leave out (but keep in the files, flagged): pseudo-element alloys (Mm, Vf: formula not defined), multi-plateau entries (one ΔH doesn't describe them), suspected entry errors, entries the Purdue curators had excluded.
4. Store with each row: the source of the value, the spread between sources, the quality flags.

Result: **725 compositions** (276 with Purdue values, 449 with HydPARK values): AB2 401, AB5 116, SS 59, AB 40, Mg 35, MIC 33, and a few small groups.

DEFEND

Q3.1 Why didn't you train on one dataset and test on the other?

Because 248 of the 288 parseable Purdue compositions are in HydPARK, mostly with identical values copied from the same papers. The test would measure memorisation of shared records, not generalisation. I used grouped cross-validation on the combined set instead.

Q3.2 Why not use ΔS as an input? It's physically related to ΔH.

In HydPARK, ΔS was back-calculated from ΔH and one (T, P) point, so it contains the answer: that's target leakage. And for a new alloy ΔS is unknown anyway. The model's purpose is to estimate ΔH from the formula alone.

Q3.3 How did you choose between conflicting values for the same alloy?

Purdue single-plateau value first (it has clear provenance and was checked against the toolbox's own code), otherwise the HydPARK median. The spread between sources is stored so it can be examined; 112 compositions with differing values have a median spread of 1.6 kJ/mol.

Q3.4 Did you delete bad data?

No. Every record is kept and flagged. Flags decide what goes into the ML union, the screening data-quality gate and the evidence class. The raw files are untouched.

Q3.5 Why exclude mischmetal alloys from the ML set but include them in screening?

Mischmetal is a variable mixture of rare earths, so its exact composition, and therefore its descriptors, cannot be computed. Screening uses only measured properties (ΔH, ΔS, capacity), which exist for those alloys, so they can be screened.

Q3.6 Is 725 compositions enough for machine learning?

It is small, which is why I used a random forest (works well on small tabular data), small hyperparameter grids, grouped and nested cross-validation, and simple baselines. It is enough to learn within-family trends (AB_5, AB_2), not enough to learn rare families. The leave-family-out test shows this.

Q3.7 How did you know H/M meant "per formula unit"?

By reading the toolbox's MATLAB loader (`ReadThermo.m`) and checking the value ranges. The Excel export's legend says "per metal atom", which is wrong.

Part 4 Turning a formula into numbers

Report: §3.4, Appendix A

A model cannot read "LaNi5". It needs numbers. This part shows exactly how a formula becomes 38 numbers, with $LaNi_5$ computed by hand, so you can do it on the board if asked.

UNDERSTAND

4.1 The idea: describe an alloy by its elements

Each element has known properties (mass, size, electronegativity …). An alloy is a mixture of elements in given fractions. So we describe the alloy by **statistics of its elements' properties, weighted by how much of each element there is**. This is the standard "composition-based descriptor" approach (Ward et al. 2016, "Magpie"). All elemental values come from **pymatgen**'s periodic table, none typed by hand.

> **THE RULE: COMPOSITION ONLY**
> Inputs come only from the chemical formula. **No ΔS, pressure, temperature, capacity or H/M** is ever used as an input (the build script asserts this). That is what makes the model usable for an alloy that has never been measured.

4.2 The 38 descriptors Report Table 3.3

Group	What	n
Elemental statistics	for 6 properties (atomic mass, electronegativity X, atomic radius, melting point T_m, molar volume, valence electrons): fraction-weighted **mean**, **standard deviation**, **min**, **max** = 24, minus 2 removed as duplicates	22
Alloy descriptors	number of elements; configurational entropy; atomic size mismatch δ; fraction of hydride-forming atoms	4
Element fractions	atomic fraction of the 12 most common elements: Zr, Ti, Mn, Fe, Ni, Cr, V, La, Co, Al, Mg, Cu	12

Pruning: if two descriptors were almost identical (|Pearson r| > 0.95), the less interpretable one was dropped. This removed the radius standard deviation (r = 0.999 with δ) and the maximum molar volume (r = 0.964 with the maximum radius). 40 − 2 = **38**.

4.3 Worked example: $LaNi_5$ by hand

STEP 1: ATOMIC FRACTIONS

$LaNi_5$ has 6 atoms per formula unit: $x_{La} = 1/6 = 0.167$, $x_{Ni} = 5/6 = 0.833$.

STEP 2: ELEMENTAL VALUES (PYMATGEN)

Element	molar volume (cm³/mol)	valence electrons	radius (Å)	hydride former?
La	22.39	3	1.95	yes
Ni	6.59	10	1.35	no

STEP 3: DESCRIPTORS

Mean molar volume (the most important descriptor): $0.167 \times 22.39 + 0.833 \times 6.59 = 3.74 + 5.49 = \mathbf{9.23}$ cm³/mol.

Valence-electron mean and spread: mean $= 0.167 \times 3 + 0.833 \times 10 = 8.83$;
std $= \sqrt{0.167(3-8.83)^2 + 0.833(10-8.83)^2} = \sqrt{5.68 + 1.14} = \mathbf{2.61}$.

Fraction of hydride formers: only La counts ⇒ **0.167**.

Configurational entropy (units of R): $-\sum x_i \ln x_i = -(0.167 \ln 0.167 + 0.833 \ln 0.833) = 0.299 + 0.152 = \mathbf{0.451}$.

Size mismatch: $\bar{r} = 0.167 \times 1.95 + 0.833 \times 1.35 = 1.450$ Å; $\delta = \sqrt{\sum x_i (1 - r_i/\bar{r})^2} = \sqrt{0.167(0.345)^2 + 0.833(0.069)^2} = \mathbf{0.154}$.

These match the project's code exactly (vol_mean 9.2286, vec_std 2.6108, config_entropy 0.4511, size_mismatch 0.1543).

4.4 Why these descriptors make physical sense

- **Mean molar volume ↔ hole size.** Bigger atoms make bigger interstitial holes, and bigger holes bind hydrogen more strongly (higher ΔH). This is the classic Lundin, Lynch and Magee (1977) correlation. In your data, vol_mean has the strongest single correlation with ΔH (Spearman 0.54, positive, as expected).
- **Fraction of hydride formers ↔ A:B ratio.** More "A" atoms (La, Ti, Zr, Mg) means more strongly bound hydrogen.
- **Valence-electron spread ↔ electronic structure.** Electron count affects how hydrogen bonds to the metal.
- **δ, configurational entropy**: lattice strain and mixing, used in multicomponent/high-entropy alloy work (Zhang et al. 2008).

A GOOD SIGN OF NO LEAKAGE

No single descriptor correlates very strongly with ΔH (maximum |Spearman| = 0.54). If one input correlated at 0.99, it would almost certainly be leaking the target.

4.5 What composition descriptors cannot see

- **Crystal structure**: the same composition can form a hexagonal C14 or a cubic C15 Laves phase with different ΔH. Descriptors are identical for both.
- **Microstructure, processing, impurities**: annealing, segregation, secondary phases.
- **Pseudo-elements**: mischmetal and ferrovanadium have undefined compositions, so they cannot be featurised.

These limits are one reason the error cannot fall to the measurement floor.

DEFEND

Q4.1 What is a "descriptor" or "feature"?

A number computed from the input that the model uses, here from the chemical formula. For example, the fraction-weighted mean molar volume of the elements.

Q4.2 Compute one descriptor for $LaNi_5$.

Mean molar volume = (1/6)(22.39) + (5/6)(6.59) = 9.23 cm³/mol. (Section 4.3 has the others.)

Q4.3 Why did you remove correlated features if random forests can handle them?

Random forests do tolerate correlation, but near-duplicates ($r > 0.95$) add nothing and split the importance between two copies, which makes interpretation harder. Ridge regression, used as a comparison, is also sensitive to collinearity.

Q4.4 Why not use DFT-calculated or structural descriptors?

Structure is known only for some compositions, and DFT for 725 alloys was beyond the project's scope. Composition-only inputs keep the model applicable to any formula. Adding structure (C14/C15, lattice parameter) is listed as the most useful future improvement.

Q4.5 You added a family one-hot encoding. Did it help?

No: with and without it the MAE is 5.26 kJ/mol within one standard deviation. The composition descriptors already encode the family, so the final model uses composition only.

Part 5 The machine-learning model

Report: §3.5 and Chapter 4, Appendix B

The part examiners probe hardest. Understand what a random forest does, why the validation is "grouped" and "nested", what the error numbers mean physically, and, most importantly, where the model stops being trustworthy.

UNDERSTAND

5.1 From one decision tree to a random forest background

A DECISION TREE IS A SEQUENCE OF YES/NO QUESTIONS

"Is the mean molar volume above 9.0?" → yes → "Is the fraction of hydride formers above 0.3?" → no → … Each question splits the training alloys into two groups. At the end of the path (a "leaf"), the prediction is the average ΔH of the training alloys that ended up there. A tree can capture non-linear effects and interactions, but a single deep tree memorises noise (it "overfits").

A **random forest** (Breiman 2001) grows many trees and averages them. Two tricks make the trees different from each other:

- **Bootstrap**: each tree trains on a random resample of the alloys (with replacement).
- **Random features**: at each split, a tree may only consider a random subset of descriptors. Your model uses **max_features = 0.33**, i.e. about a third of the 38 at each split.

Averaging 300 different, individually overfitting trees cancels much of their noise. Your final settings: **300 trees, max_features 0.33, min_samples_leaf 1, random_state 0** (the seed, so results repeat exactly).

WHY A RANDOM FOREST AND NOT A NEURAL NETWORK?

725 rows of tabular data is small. Tree ensembles are the standard strong choice for small tabular data: non-linear, no feature scaling needed, little tuning, and easy to inspect (permutation importance). A neural network would need far more data and would be harder to validate honestly. Gradient boosting (another tree ensemble) was also tested and performed the same.

5.2 How good is "good"? Metrics and baselines

MAE

Mean absolute error: average of |true − predicted|. In kJ/mol. "On average the prediction is off by this much."

RMSE

Root-mean-square error. Squares the errors first, so a few **large** errors raise it a lot. RMSE ≫ MAE means "a few bad outliers".

R^2

Share of the variation in ΔH explained. 1 = perfect, 0 = no better than always predicting the mean.

Baselines are the "dumb" predictors a model must beat to be worth anything:

- **Mean baseline**: always predict the average ΔH of the training set. MAE 12.26.
- **Family-median baseline**: predict the median ΔH of the alloy's family (AB_5, AB_2 …). MAE 8.88. This is what a hydride chemist's rule of thumb would give.

5.3 Cross-validation: grouped and nested Report §3.5

k-fold cross-validation: split the data into 5 parts; train on 4, test on the 5th; rotate so every part is tested once; average. Repeat with 3 different shuffles. Every alloy is predicted by a model that never saw it.

WHY "GROUPED"?

The data contain near-duplicates, e.g. $LaNi_5$ and $LaNi_{4.95}Sn_{0.05}$ (L1 distance 0.016). If one is in training and the other in testing, the test is too easy. So compositions with the **same elements and an L1 distance ≤ 0.02** are put in the same group (725 compositions → 659 groups), and a whole group always goes to the same side of the split. Effect: ungrouped CV would have reported MAE 5.12, grouped gives 5.26, about 3% more honest.

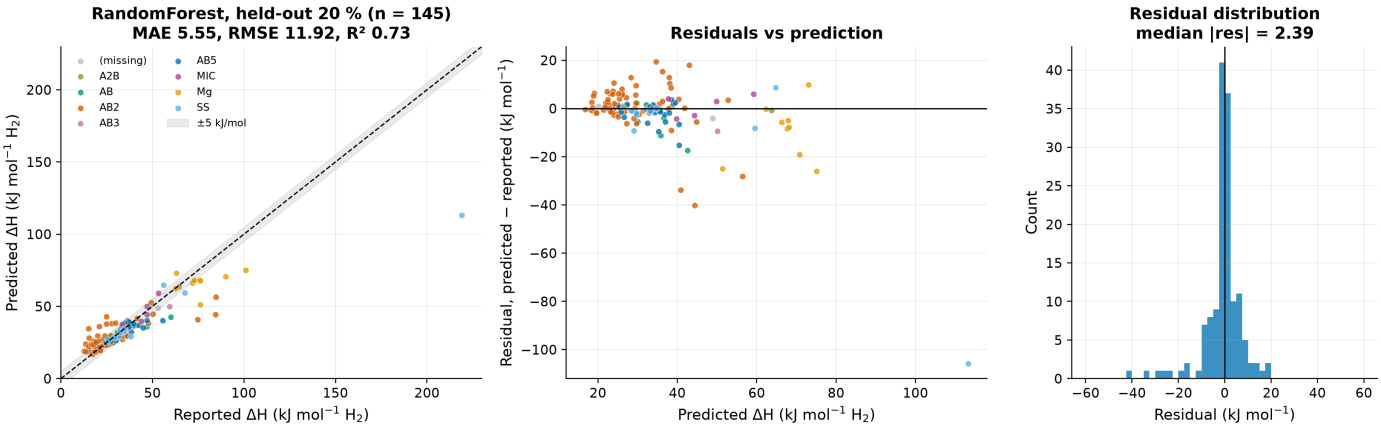
WHY "NESTED"?

Hyperparameters (e.g. max_features) are chosen by trying a few values and picking the best. If you pick using the *same* test folds you then report, the reported error is optimistically biased (Varma & Simon 2006). Nested CV has two loops: the **inner** loop (4-fold, inside each training part only) picks the hyperparameters; the **outer** loop (5-fold × 3 repeats) measures the error on data never used for any choice.

5.4 The results Report Table 4.2

Model	MAE (kJ/mol)	RMSE	R^2
Random forest (chosen)	**5.26 ± 0.14**	10.32	0.715
Gradient boosting	5.30 ± 0.13	9.77	0.744
Ridge (linear)	7.09	12.3	0.60
Family-median baseline	8.88	16.3	0.29
Mean baseline	12.26	19.4	0.00

- RF is **41% better** than the family rule and 57% better than the mean.
- RF and GB are the same within noise; RF was kept (not better or worse on MAE, the main metric).
- Ridge is clearly worse → the composition–ΔH relation is **non-linear**.
- RMSE ≈ 2 × MAE because a few very stable hydrides (ΔH > 60) have large errors.
- **Hold-out check**: on one separate grouped 20% split (145 alloys), RF MAE 5.55, R^2 0.73. Consistent.



Hold-out parity and residuals (report Fig. 4.2). Points close to the diagonal are good predictions.

5.5 What 5.26 kJ/mol means physically

- **Against the data floor**: independent measurements of the same alloy differ by ≈ 2.2 kJ/mol. The model's error is ≈ 2.4× that. No model can go much below ≈ 2 with these data.
- **Against the operating point**: 5 kJ/mol ≈ a factor 7.5 in plateau pressure at 25 °C. So the model can tell a 25 kJ/mol alloy (several bar at room temperature) from a 40 kJ/mol alloy (needs ≈ 100 °C), but it cannot design an exact plateau pressure. **It is a screening tool.**
- **Data quality**: on a cleaner subset of 456 compositions (referenced, typical ΔS, metals only), MAE drops to 3.51. Part of the error is the data, not the model.

5.6 By family, and the leave-one-family-out test Report §4.4-4.5

Within the big families the model is good: **AB_5 MAE 3.09** (family rule 5.15), **AB_2 4.17** (7.12). So it learns substitution effects, like Al or Sn replacing Ni in $LaNi_5$.

The hard test: remove a whole family from training, then predict it.

Held-out family	n	Mean of other families	Random forest	Verdict
AB_5	116	5.10	5.86	worse than a constant
AB_2	401	17.50	13.84	better, still large
AB	40	16.21	16.52	worse
SS	59	12.33	20.98	much worse
Mg	35	36.01	14.51	better, still large
MIC	33	14.30	11.09	better, still large

THE HONEST CONCLUSION

The model does not generalise to an unseen family. It interpolates within and near known families. That is why it is never used to screen Mg-based or complex hydrides, and why every prediction carries a domain flag. Saying this openly is a strength in a viva, not a weakness.

5.7 Applicability domain and the 80% interval Report §4.6

Distance to the training data. For a query alloy, compute the **L1 distance** between its fraction vector and every training alloy's, and take the smallest, d_{NN}.

EXAMPLE: $LaNi_{4.2}Al_{0.8}$ VS THE NEAREST KNOWN ALLOY $LaNi_{4.25}Al_{0.75}$

Fractions (La, Ni, Al): (0.167, 0.700, 0.133) vs (0.167, 0.708, 0.125). $d_{NN} = |0.167 - 0.167| + |0.700 - 0.708| + |0.133 - 0.125| = 0 + 0.008 + 0.008 = 0.016$.

Calibration. For every out-of-fold prediction, the distance to its nearest training alloy (outside its own group) was recorded, and errors were binned by distance:

d_{NN}	≤ 0.05	0.05-0.10	0.10-0.15	0.15-0.20	0.20-0.30	0.30-0.50	> 0.50
n	219	220	94	86	27	57	22
MAE	2.62	4.22	4.63	6.14	6.88	12.9	16.9
q10 / q90 of residual	−2.95 / 4.34	−5.76 / 6.36	−7.04 / 7.23	−7.81 / 10.7	−13.8 / 5.67	−32.2 / 8.28	−22.3 / 23.7

HOW THE 80% INTERVAL IS COMPUTED (RESIDUAL = PREDICTED − TRUE)

$$[\,\hat{y} - q_{90},\ \hat{y} - q_{10}\,]$$

$LaNi_{4.2}Al_{0.8}$: $\hat{y} = 38.69$, bin ≤ 0.05 → $[38.69 - 4.34,\ 38.69 + 2.95] = [34.3,\ 41.6]$ kJ/mol. ✓ (in domain)
$Ti_{0.5}V_{0.5}$: $d_{NN} = 0.333$, bin 0.30–0.50 → $\hat{y} = 54.4$, interval $[46.1,\ 86.6]$; bin MAE 12.9 → **flagged OUTSIDE DOMAIN**.

- **Domain limit $d_{NN} \le 0.15$**: beyond it the typical error exceeds 6 kJ/mol and climbs fast.
- **Coverage check**: on 22 alloys of four substitution series with known values, 91% of the measured ΔH fell inside the nominal 80% interval (mean error 1.91 kJ/mol).
- The interval is **empirical**: the typical spread of past errors at that distance. It is not a guarantee for one alloy.

5.8 What the model uses: permutation importance Report §4.7

Method: take the hold-out set, shuffle one descriptor's column randomly, and see how much the error increases. A big increase means the model relies on that descriptor.

Top three: **mean molar volume** (+3.57 kJ/mol), **fraction of hydride formers** (+2.10), **valence-electron spread** (+1.31). The first matches the hole-size rule (Lundin 1977) and the volume descriptor Witman et al. (2020) found on HydPARK. Physically sensible. But descriptors are correlated, so importance is shared: this explains the model, not causation.

5.9 The capacity model that was not used

A second model tried to predict wt% (181 Purdue records with a stated wt%). Best MAE 0.247 wt%, only slightly better than a **physics baseline** (0.286 wt%: take the family's typical H per metal atom and divide by the alloy's mass). The model mostly rediscovered "heavier alloys hold less wt%". With narrow ranges and inconsistent definitions, it was **reported but not used**. Screening uses reported capacities.

5.10 The deployed model

The final RF was refitted on all 725 compositions with the same settings, then packaged **unchanged** into a bundle with the training compositions, the calibration table and the ΔS assumption. A manifest records the **SHA-256 hash** of each file (a fingerprint that changes if a single byte changes), so you can prove the app uses exactly the validated model. It was never retrained after validation.

DEFEND

Q5.1 R^2 is only 0.72. Isn't that weak?

Judge it against the baselines and the data. The family rule gets R^2 0.29, and independent measurements of the same alloy already scatter by ≈ 2.2 kJ/mol. R^2 is also pulled down by a few very stable outliers. Within AB_5/AB_2 the MAE is 3–4 kJ/mol, and on clean data 3.5. For screening that is useful; for design it is not, and I say so.

Q5.2 What is overfitting and how did you avoid it?

A model that memorises training noise and fails on new data. Controlled by averaging many trees, small hyperparameter grids, and above all by measuring error only on data the model never saw (nested, grouped CV, plus a separate hold-out).

Q5.3 Why 5 folds and 3 repeats?

Five folds leave 80% for training, which matters with 725 rows; three repeats with different shuffles show how stable the estimate is (± 0.14 kJ/mol). The inner loop uses 4 folds inside each training part.

Q5.4 Why is the interval asymmetric for $Ti_{0.5}V_{0.5}$?

Because it uses the actual error distribution of that distance bin. In that bin some alloys were badly *under*-predicted (10% residual quantile −32.2 kJ/mol), so the interval stretches upward.

Q5.5 Why is the domain limit 0.15 and not something else?

It comes from the calibration: up to 0.15 the MAE stays below 5 kJ/mol (2.6, 4.2, 4.6); just beyond it is 6.1, and after 0.3 it jumps to 13–17. 0.15 is where the error starts to exceed the useful screening resolution. It is a judgement based on that table.

Q5.6 Is the 80% interval a confidence interval?

Not in the strict statistical sense. It is an empirical prediction interval: the 10–90% range of past out-of-fold errors for alloys at a similar distance from the training data. It assumes new alloys behave like past ones at that distance. On a check set, 91% of true values fell inside.

Q5.7 What does "out-of-fold" mean?

A prediction for an alloy made by a model trained on the other folds, i.e. a model that never saw that alloy. All reported errors and the calibration use out-of-fold predictions.

Q5.8 If I type $LaNi_5$ into your app it says 31.4, but the measured value is 32.15. Why?

LaNi5 is in the training data, so 31.4 is an in-sample value (the app warns about this). The honest out-of-fold prediction, from a model that never saw LaNi5, was 29.4. For training alloys, always quote the measured value.

Part 6 Screening and ranking materials

Report: §6.1–6.2, §7.3–7.5, Appendices D–E

How the project turns "here are 337 measured alloys" into "for this application, these are the best candidates, and here is why". Three ideas: hard constraints (pass/fail), a weighted score (ranking), and the Pareto front (trade-offs). Then a sensitivity test to see whether the ranking is robust.

UNDERSTAND

6.1 The principles

1. **Experimental data first.** Rankings use the 337 Purdue materials with *measured* ΔH, ΔS. ML predictions are shown separately and never mixed into this ranking.
2. **Material level only.** Capacities are reported material values; no tank or system design.
3. **Missing is not failing.** Each constraint gives **pass**, **fail**, **unknown** (data missing) or **not specified**. A material is rejected only on a real failure.
4. **Everything visible.** Constraints, criteria and weights are printed with the results and tested for sensitivity.

6.2 Step 1: hard constraints Report Table 6.1

A hard constraint is a yes/no engineering limit. First a **data-quality gate** (ΔS between 70 and 170 J/mol·K; capacity not "assumed" or "unconfirmed"), then the application's limits:

Application	Constraint	Engineering reason
Mobile	T(1 bar, full capacity) ≤ 80 °C	PEM fuel-cell waste heat is at 60–80 °C
Mobile	P(50 °C) ≤ 50 bar	tank pressure on a hot day
Mobile	charging P(25 °C) ≤ 100 bar	refuel without very high compression
Stationary	T(1 bar) ≤ 150 °C	low-grade industrial heat
Stationary	P(25 °C) ≤ 10 bar; P(50 °C) ≤ 30 bar	low-pressure, cheaper, safer vessel
Stationary	charging P(25 °C) ≤ 30 bar	fill directly from a PEM electrolyser (30 bar assumed)

Outcome: 319 of 337 pass the gate; **232** pass all mobile constraints, **252** all stationary. Mobile mainly rejects alloys needing too much heat (56) or too much pressure on a hot day (35); stationary mainly rejects high-pressure alloys (57 at 25 °C). No Mg-based material passes either.

6.3 Step 2: the weighted score

Each criterion is first scaled to 0–1 among the materials that passed (min–max normalisation):

$$s = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \ \text{(higher is better)}, \qquad s = 1 - \frac{x - x_{\min}}{x_{\max} - x_{\min}} \ \text{(lower is better)}$$

then combined: score $= \sum_k w_k s_k$ with weights summing to 1.

Criterion	Mobile	Stationary
capacity	0.35	0.10
reaction heat	0.20	0.10
kinetics class	0.15	0.10
hysteresis	0.10	0.15
pressure fit	0.10	0.10
cost class	0.10	0.25
pressure safety	–	0.20

Kinetics class is family-level: AB_5/AB_2/AB/AB_3 = 1, complex = 0.25, Mg = 0, no data = 0.5. Individual kinetic data span 7 orders of magnitude and depend on each lab's setup, so they are not comparable alloy by alloy.

Cost class is an element indicator, *not a price*: common metals 1; rare earths, Co, V, minor metals 0.5; precious metals 0.

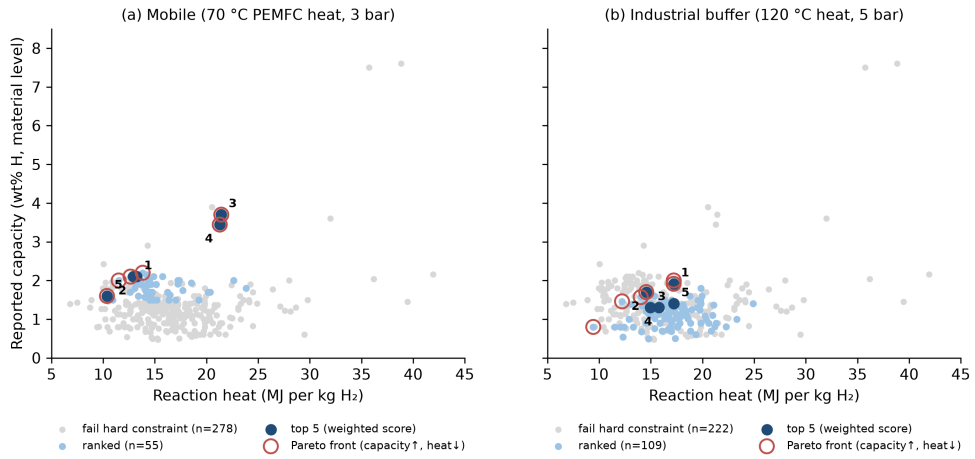
Missing hysteresis → 0.5 (neutral) and flagged.

Mobile weights favour capacity and low heat; stationary weights favour cost, low pressure and low cycling loss, because weight barely matters for a fixed store.

6.4 Step 3: the Pareto front

> **DOMINANCE IN ONE SENTENCE**
>
> Material A **dominates** B if A is at least as good on every objective and strictly better on one. The **Pareto front** is the set of materials that nobody dominates: the best possible trade-offs.

With objectives "capacity ↑" and "reaction heat ↓": in the mobile scenario, $V_{0.855}Ti_{0.095}Fe_{0.05}$ (3.7 wt%, 21.4 MJ/kg) and $TiCrMn_{0.85}Fe_{0.3}V_{0.15}$ (1.6 wt%, 10.4 MJ/kg) are both on the front: one has more capacity, the other needs half the heat. Neither is "better"; the user chooses the trade-off. The weighted score gives one ranking; the Pareto front shows the trade-off without choosing weights.



Two scenarios: grey = fails a hard constraint, light blue = ranked, dark = top 5, red circles = Pareto front (report Fig. 7.1).

6.5 Step 4: is the ranking robust? Report §6.2

- **Random weights**: 2000 weight sets drawn from a **Dirichlet distribution** (a way to draw random positive weights that sum to 1) centred on the chosen weights, plus 2000 completely random ones. Result: the leading mobile alloy stays in the top 10 in 88% (near the chosen weights) and 72% (any weights) of draws.
- **One-at-a-time**: set each weight to 0 or double it. Ranking agreement is measured with **Kendall τ** (1 = same order, 0 = unrelated). Largest effects: removing capacity for mobile ($\tau = 0.71$), doubling pressure safety or removing cost for stationary ($\tau = 0.65$).
- **Thresholds**: e.g. mobile T(1 bar) limit of 60, 80 or 100 °C → 203, 232, 253 pass; the top 3 stay the same.

> **HOW TO DESCRIBE THE RESULT**
>
> "The **membership** of the top group is robust: Ti-based AB_2/AB for mobile, TiFe-type and Ti–Zr–Mn for stationary. The exact **order** inside the group depends on the weights, so I read the ranking as tiers, not as one winner."

6.6 Evidence classes

Class	Meaning	Ranked?
Data-supported	measured ΔH, ΔS and a literature capacity; no caveats	yes
Provisional	passes the gate but has a caveat: own PCT curves disagree with the tabulated ΔH by > 5 kJ/mol, unverified Sandia reference, data identical to another file, pseudo-element, or approximate capacity	yes, flagged
Insufficient evidence	fails the data-quality gate	shown, **never ranked**

6.7 The results you should know Report §7.3-7.4

Generic mobile screen, top 5

1. $Ti_{1.3}Cr_{1.2}Mn_{0.8}$ (AB_2, 1.90 wt%, 18.6 bar at 25 °C)
2. $Ti_{1.2}CrMn$ (AB_2, 2.02 wt%)
3. $Ti_{0.56}V_{0.3}Zr_{0.1}Mn_{0.94}Fe_{0.06}Ni_{0.04}$ (AB) † its own PCT curves disagree
4. $Ti_{0.9}Zr_{0.2}Mn_{1.4}Cr_{0.4}V_{0.2}$ (AB_2) ‡ contains ferrovanadium
5. $Ti_{0.8}Zr_{0.2}Mn_{1.5}Cr_{0.5}$ (AB_2) § unverified reference

Generic stationary screen, top 5

1. $TiFe_{0.9}Al_{0.1}$ (AB, 1.41 bar at 25 °C)
2. $Ti_{0.9}Zr_{0.1}Mn_{1.4}$ (AB_2, 0.80 bar)
3. $Ti_{0.5}Zr_{0.5}Mn_{1.2}Fe_{0.3}$ (AB_2) §
4. $TiFe_{0.8}Mn_{0.2}$ (AB)
5. $TiMn_{1.4}$ (AB_2) †

$LaNi_5$, the textbook reference, passes both but ranks only 58th (mobile) and 66th (stationary): modest capacity (1.49 wt%) and the rare-earth cost class.

Two demonstration scenarios (illustrative requirements, run in the platform):

	Mobile (PEMFC vehicle)	Industrial buffer
Key inputs	70 °C waste heat, 5 kW; 1 kg H_2; deliver 3 bar; ≤ 80 kg alloy; ≥ 1.5 wt%	120 °C waste heat, 50 kW; 50 kg H_2; deliver 5 bar; ≤ 30 bar
Fail / unranked / ranked	278 / 4 / 55	222 / 6 / 109
Top 3	$Ti_{0.9}Zr_{0.2}Mn_{1.4}Cr_{0.4}V_{0.2}$, $Ti_{0.9}Zr_{0.2}Mn_{1.8}V_{0.2}$, $V_{0.855}Ti_{0.095}Fe_{0.05}$	$Ti_{0.9}Zr_{0.1}Mn_{1.4}$, $Zr_{0.7}Ti_{0.3}CrFe$, $TiFe_{0.9}Al_{0.1}$
Telling numbers	top alloy: ≈ 48 kg for 1 kg H_2, 1.8 kW heat	top alloy: ≈ 2.6 t for 50 kg H_2; needs ≈ 70 °C, 120 °C available

The two top-5 lists share **zero** materials. When each of the seven weights is removed or doubled (14 cases), the industrial leader stays first in 9, the mobile leader in only 4: **mobile rankings depend much more on priorities**.

DEFEND

Q6.1 How did you choose the weights?

From the engineering priorities of each application: capacity and heat matter most for a vehicle; cost, low pressure and cycling loss for a stationary store. Because weights are a judgement, I tested them: 4000 random weight sets and one-at-a-time changes. The top group stays the same; the order within it changes.

Q6.2 Why use hard constraints and a score, not just a score?

Some requirements are absolute: an alloy that cannot deliver 3 bar with 70 °C heat is useless however good its capacity. A score could hide that. So constraints remove impossible materials, and the score ranks only feasible ones.

Q6.3 What is a Pareto front, and why show it as well as the ranking?

The set of materials not beaten on all objectives at once. It shows the real trade-offs (capacity vs heat) without imposing weights. The ranking needs weights; the front does not.

Q6.4 Your top mobile alloy contains vanadium, which you class as costly. Contradiction?

No. Cost has a weight of 0.1 for mobile, where capacity and heat dominate. With the stationary weights (cost 0.25) V-containing alloys drop. The cost class is qualitative, not a price model.

Q6.5 Is your #1 material the best hydride for vehicles?

It is the top result of a screening of literature data under stated requirements and weights. Several top entries carry data caveats, and any candidate needs its own PCT measurement, cycling and kinetics tests before use. I present the result as a short list to measure.

TRAP "Best" is an overclaim; "top-ranked under these requirements" is correct.

Q6.6 What happens to a material with missing hysteresis data?

It is not rejected. The hysteresis criterion gets a neutral 0.5 and the material is flagged; for the charging-pressure check, the desorption pressure is used and marked "optimistic".

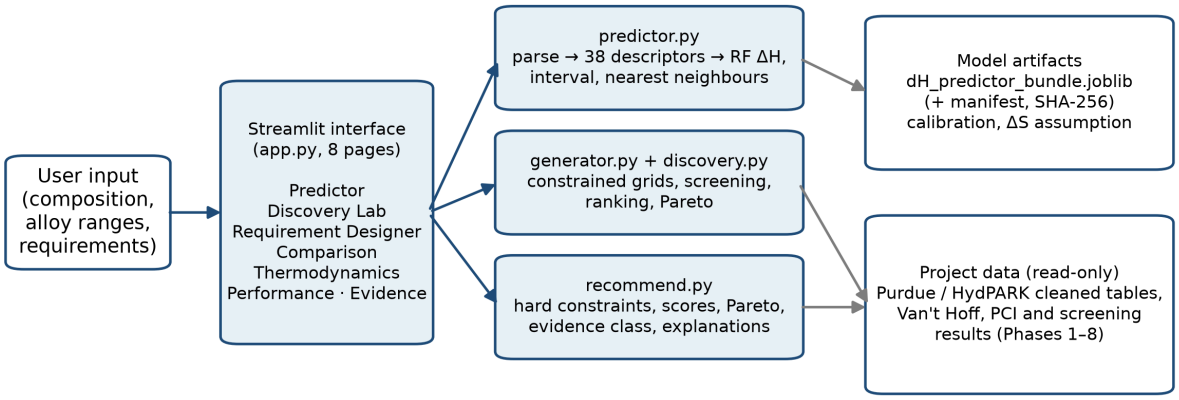
Part 7 The software platform

Report: §6.3; code in `app.py` and `src/app/`

The app ("METAL HYDRIDE AI") is the visible part of the project. Know what each page does, what happens under the hood when you click, and have a short demo ready.

UNDERSTAND

7.1 Architecture



Pages (Streamlit) call engine modules; engines read the model bundle and project data read-only (report Fig. 6.2).

The app computes nothing of its own. Every ΔH comes live from the saved random forest, and every thermodynamic number from the same Van't Hoff functions used in the report. The engines contain no user-interface code, so they can be tested without a browser.

7.2 The eight pages

Page	What it does
Overview	model metrics read from the result files, dataset coverage, shortcuts
Property Predictor	type a formula → ΔH, 80% interval, domain flag, 5 nearest measured alloys, labelled Van't Hoff estimate
Candidate Discovery Lab	generate a grid of compositions in an alloy system → predict → reject with reasons → rank → Pareto front
Application Requirement Designer	7-step requirement form → hard constraints → ranking, Pareto, evidence, explanation, export
Material Comparison	measured properties side by side, with provenance; missing shown as missing
Thermodynamic Explorer	Van't Hoff curves with ΔS sensitivity and neighbour lines
Model Performance	the validation results of Part 5
Dataset & Evidence Explorer	overlap, flags and missing data of Part 3

7.3 What happens when you press "Predict"

1. **Parse** the text (e.g. "LaNi4.2Al0.8") into elements and amounts; reject invalid input with a message.
2. **Normalise** to atomic fractions and build the composition key.
3. **Featurise**: compute the 38 descriptors (Part 4), using the same code as in training.
4. **Predict** ΔH with the saved random forest → 38.69 kJ/mol.

5. **Distance**: L1 distance to all 725 training alloys → nearest 0.016.
6. **Interval**: look up the calibration bin → [34.3, 41.6]; domain check (≤ 0.15) → in domain.
7. **Warnings**: unseen element? already in training (then the value is in-sample)? outside domain?
8. **Van't Hoff estimate** with the assumed family-median ΔS, labelled as an estimate.

7.4 Discovery Lab and Requirement Designer

Discovery Lab: choose a template ($TiFe_{1-x}Mn_x$; Ti–Zr–Mn–Cr or Ti–Cr–Mn Laves; La–Ni–Al, La–Ni–Sn, La–Ce–Ni–Al AB_5; V–Ti–Cr BCC) or free fractions, set ranges and step. The generator builds a reproducible grid and removes duplicates. A candidate is **rejected** if it has unseen elements, lies outside the domain, or if even the optimistic end of its temperature band misses the target. Survivors are ranked. All outputs say: ΔH predicted, ΔS assumed, capacity not predicted, candidates only.

Requirement Designer: Step 1 application type (5 presets), 2 environment (ambient range, heat source), 3 hydrogen needs (kg, delivery pressure, flow), 4 optional limits (mass, volume, pressures, heating power), 5 weights, 6 review, 7 results. It derives the operating conditions: delivery temperature = heat-source temperature if any, else the coldest ambient; hottest bed temperature; charging temperature (cooling water or hottest ambient). Then it evaluates every measured material.

7.5 Testing, and a bug found while writing the report

77 automated tests check parsing, consistency with the training pipeline, the generator, constraints, scoring, Pareto fronts, evidence classes, export and a headless run of every page. While writing the report, reviewing the demo output showed that unranked "insufficient evidence" materials were included in the Pareto computation and could dominate the ranked ones. In the industrial scenario this made the front among ranked materials appear empty. The fix computes the front among ranked materials only, and a new test guards it. Rankings did not change.

7.6 A five-minute demo for the viva

RUN

`cd Hydrogen-Storage-Project` then `.venv/bin/streamlit run app.py`, which opens `http://localhost:8501`. Open it before the viva and keep it running.

1. **Overview** (20 s): "725 compositions, MAE 5.26, read live from the result files."
2. **Property Predictor**, type `LaNi4.2Al0.8`: "38.7 kJ/mol, interval 34.3–41.6, in domain, nearest known alloy $LaNi_{4.25}Al_{0.75}$ at 38.3."
3. Type `Ti0.5V0.5`: "outside the domain, distance 0.33, typical error 13 kJ/mol. The app refuses to pretend."
4. Type `LaNi5`: "it warns this is a training alloy, so the measured 32.15 is the value to quote."
5. **Requirement Designer**, choose "Mobile hydrogen storage" and run: "every rejected material lists the requirement it fails; the ranked ones show an evidence class; the Pareto front shows capacity against heat." (The report's scenario 1 adds a mass limit of 80 kg and a minimum of 1.5 wt%: 278 rejected, 55 ranked.)
6. **Model Performance**: "leave-family-out: the model fails on unseen families, and that is why the domain flag exists."

DEFEND

Q7.1 Does the app use the same model as the report?

Yes, the identical saved random forest, packaged unchanged. A manifest stores SHA-256 hashes of the model files, and a test checks that the app's predictions match the training pipeline.

Q7.2 Why Streamlit?

It turns Python analysis code into an interactive web app quickly, so the same tested engine functions serve the report and the app. For a research prototype that matters more than a custom front end.

Q7.3 How do you know the app's numbers are right?

The engines are separate from the interface and covered by 77 tests, including consistency between the app predictor and the training pipeline. All thermodynamics use the same Van't Hoff functions as the report, and the demo outputs were regenerated and compared after the Pareto fix.

Q7.4 What stops a user from trusting a bad prediction?

Every prediction shows its interval and the distance to the nearest known alloy; outside the domain it is labelled OUTSIDE DOMAIN with the typical error at that distance; training alloys are flagged as in-sample; unseen elements trigger a warning; Lab outputs are always labelled as candidates.

Part 8 Limitations and future work

Report: §7.6, Chapter 8, Appendix F

Examiners respect a student who knows exactly where their work stops. Know each limitation, what it does to the results, and what would fix it. Then know the proposed experimental programme well enough to explain each technique in one sentence.

UNDERSTAND

8.1 The limitations, honestly Report Table 7.4

Limitation	Consequence	Fix
No independent test set	CV error is the only estimate of generalisation	newly measured alloys (blind test)
Composition-only descriptors	can't see C14 vs C15, microstructure, processing	XRD phase data; structure descriptors
Fails on unseen families	no ML screening of Mg, complex or new families	data for those families; physics-informed models
ΔS assumed, not predicted	T(1 bar) bands ≈ 105 K for new alloys	PCT isotherms at ≥ 4 temperatures; a ΔS model
Capacities: unclear definition, material level	capacity rankings indicative; no system capacity	reversible capacity between the application's pressures; a system model
Density for 37 materials only	volumetric capacity mostly unknown	density, bed packing measurements
Kinetics and cost qualitative	no rate or price optimisation	standardised kinetic tests; cost data
No cycle-life data	durability not screened	≥ 100-cycle tests
Digitised isotherms	PCT check is only a consistency check	original measured isotherms
No laboratory work	all rankings remain screening results	the proposed programme below

8.2 The proposed experimental programme, in plain words Report §8.3

Nothing here was done. It is a plan. **Nine alloys** from the screening: $LaNi_5$ and TiFe as *benchmarks* (known behaviour, to check the equipment); $Ti_{1.2}CrMn$, $Ti_{0.9}Zr_{0.2}Mn_{1.4}Cr_{0.4}V_{0.2}$ and $Ti_{0.56}V_{0.3}Zr_{0.1}Mn_{0.94}Fe_{0.06}Ni_{0.04}$ from the mobile tier (the last because its data conflict); $TiFe_{0.9}Al_{0.1}$, $Ti_{0.9}Zr_{0.1}Mn_{1.4}$, $TiFe_{0.85}Mn_{0.15}$ and $CaNi_5$ from the stationary tier. Plus **two blind tests of the model**: $LaNi_{4.2}Al_{0.8}$ (predicted 38.7, interval 34.3–41.6) and $TiFe_{0.75}Mn_{0.25}$ (32.0, 25.7–37.8). The model "passes" if the measured ΔH falls in the interval.

Technique	What it does, in one sentence
Arc melting (argon, Ti getter, 4–5 remelts)	melts the pure metals together with an electric arc in inert gas, flipping and remelting so the alloy is uniform
Homogenisation anneal	holds the alloy hot in a sealed tube so segregated regions even out, then quenches
ICP-OES	dissolves a sample and measures the actual element amounts (is it really $LaNi_{4.2}Al_{0.8}$?)
XRD + Rietveld refinement	X-ray diffraction identifies the crystal phases (C14 vs C15) and lattice parameters
SEM/EDS	electron microscope images of microstructure plus local composition; shows secondary phases
Activation	first few hydrogen cycles (with heating) to crack surface oxides; essential for TiFe
Sieverts PCT, abs + des, ≥ 4 temperatures	the key test: measures plateau pressure vs hydrogen content at several temperatures → ΔH, ΔS, capacity, slope, hysteresis
Kinetic curves	uptake/release vs time at fixed pressure ratio and several temperatures → rates, activation energy

Cycling (≥ 100 cycles)	how much capacity is lost over time
DSC	measures reaction heat directly; an independent check of ΔH
Thermal conductivity, bed density	needed for heat-exchanger design and volumetric capacity

8.3 Future modelling and platform work Report §8.4

- Add structural information (C14/C15, lattice parameters).
- Model ΔS, or the whole Van't Hoff line, instead of assuming it.
- **Active learning**: measure the candidates the model is least sure about, add them, retrain, and the domain grows where experiments are made.
- A **system-level model** (vessel, heat exchanger, porosity) for system capacity and refuelling time.
- Quantitative cost and kinetic data where available.

DEFEND

Q8.1 What is the weakest part of your project?

The absence of an independent test: the two datasets share most values, and no new alloys were measured. The cross-validation is done carefully (grouped, nested), but a blind test on new alloys would be the real proof. That is why two blind-test compositions with stated intervals are proposed.

Q8.2 If you had six more months, what would you do first?

Synthesise and PCT-test the two blind-test alloys and the benchmarks, to check the model against new measurements. Then add XRD phase information as a descriptor, and build a ΔS model so the temperature bands shrink.

Q8.3 Why choose $LaNi_5$ and TiFe for the experiments when you already know their values?

They are benchmarks. If my set-up reproduces their known plateaus, I can trust its measurements on the new alloys.

Q8.4 What is a Sieverts apparatus?

A volumetric set-up: a calibrated reservoir of hydrogen at known pressure and temperature is opened to the sample; the pressure drop tells how much hydrogen was absorbed. Repeating step by step traces the PCT curve.

Part 9 Viva question bank

More questions, mixed across the project, in the order an examiner tends to escalate: basic → method → critical → traps. Answer in two or three sentences, start with the direct answer, then give one number as evidence. If you don't know, say what you would check.

DEFEND Level 1: basic understanding

B1 What is the objective of your project?

To go from an alloy's chemical formula to a defensible statement about whether it suits a given hydrogen-storage application: a clean dataset, a ΔH model with known reliability, Van't Hoff operating windows, and a requirement-driven screening tool.

B2 Why is ΔH the target, not capacity?

ΔH decides the operating temperature and heat demand, which decide whether an alloy is usable at all. Capacity data are fewer and inconsistently defined; a capacity model was tested and barely beat a physics baseline (0.247 vs 0.286 wt%), so it is not used.

B3 What is a metal hydride, in one line?

A metal or alloy that absorbs hydrogen atoms into gaps in its crystal lattice, reversibly, releasing heat on charging and needing heat on discharging.

B4 What data did you use?

Two public literature compilations: the Purdue Metal Hydride Toolbox (337 materials) and Sandia's ML-HydPARK v0.0.5 (772 rows). After audit and merging: 725 compositions with measured ΔH.

B5 What software and libraries?

Python: pandas for data, pymatgen for composition parsing and elemental data, scikit-learn for the models and cross-validation, matplotlib for figures, Streamlit for the app, pytest for the 77 tests.

B6 What is the DOE target and how do you compare?

The DOE 2025 onboard target is 5.5 wt% for the whole *system* (1.8 kWh/kg). Interstitial hydrides reach ≈ 1–2 wt% as a *material*; none comes close. Hence hydrides for stationary or heavy-duty uses.

B7 What is the unit of ΔH and why "per mol H_2"?

kJ per mol of H2 released. Normalising per H2 lets alloys with different formulas and capacities be compared directly, and it is the quantity that enters Van't Hoff.

B8 What does "applicability domain" mean?

The region of composition space where the model's predictions can be trusted, here alloys within L1 distance 0.15 of a training alloy, where the typical error is below about 5 kJ/mol.

DEFEND Level 2: methodology

M1 Walk me through your validation.

Group near-identical compositions (same elements, L1 ≤ 0.02 → 659 groups). Outer 5-fold grouped CV repeated 3 times measures error; inside each training part a 4-fold grouped grid search chooses hyperparameters. Compare against mean and family-median baselines. Then a separate grouped 20% hold-out, leave-one-family-out tests, and data-subset refits.

M2 Why compare with baselines at all?

An error number means nothing alone. A chemist's family rule already gets 8.88 kJ/mol; the model's value is the 41% improvement over that, to 5.26.

M3 **What hyperparameters did you tune, and why so few?**

RF: max_features {0.33, 0.66}, min_samples_leaf {1, 3}; GB: learning rate and depth; ridge: α. With 725 rows a big search mostly fits noise in the inner folds. Selected RF: 300 trees, 0.33, 1.

M4 **How did you get the 80% interval?**

From out-of-fold residuals binned by distance to the nearest training alloy: interval = prediction minus the 90% and 10% residual quantiles of that bin. For $LaNi_{4.2}Al_{0.8}$: 38.7, interval 34.3–41.6.

M5 **Why L1 distance and not Euclidean?**

L1 on atomic fractions has a direct meaning: the total fraction of atoms you would have to change to turn one composition into the other (counted on both sides). It is simple, interpretable, and calibrated against the actual errors, which is what matters.

M6 **How did you merge duplicates from the two databases?**

One row per composition key: Purdue single-plateau value if present, else the HydPARK median. Pseudo-elements, multi-plateau entries and suspected errors excluded but kept and flagged.

M7 **How did you check your thermodynamic code?**

Against the toolbox's own MATLAB function (same equation), against HydPARK's ln P(25 °C) column, by hand (the $LaNi_5$ example), and against the digitised isotherms (median ΔH difference 1.76 kJ/mol; $NaAlH_4$ plateau assignment confirmed).

M8 **Why min-max normalisation of criteria?**

It puts criteria with different units (wt%, MJ/kg, ratios) on a common 0–1 scale among the feasible materials, so weights mean "relative importance". Its weakness is sensitivity to extreme values, one reason the sensitivity analysis was done.

M9 **What is Kendall τ?**

A rank-correlation measure: the fraction of pairs ordered the same way minus the fraction ordered differently. 1 = identical ranking. I used it to compare rankings after changing one weight.

DEFEND Level 3: critical questions

C1 **Your model fails on unseen families. So what is it good for?**

Exploring substitutions within known families, which is how most hydride alloys are actually developed (Al/Sn in $LaNi_5$, Mn in TiFe, Ti/Zr and Cr/Mn ratios in Laves phases). There it reaches 3–4 kJ/mol, and it tells you when you have left that region.

C2 **Isn't the dataset biased towards AB_2 and AB_5?**

Yes: AB_2 is 401 of 725 compositions. That is why errors are reported per family, why leave-family-out was run, and why predictions for sparse families are flagged. The bias reflects the literature.

C3 **The error is 2.4× the measurement scatter. Could a better model close the gap?**

Partly. On the high-confidence subset the MAE is already 3.51, so data quality explains part of the gap. The rest is likely missing information (crystal structure, processing), which composition descriptors cannot supply. Gradient boosting gave the same MAE, so the model type is not the bottleneck.

C4 **Are your rankings just a product of your weights?**

The order within the top group is, which I state. The membership is not: across 4000 random weight sets the same families dominate, and the top mobile alloy stays in the top 10 in 72% of completely random weightings.

C5 **You treat kinetics at family level. Isn't that crude?**

Yes, deliberately. The individual kinetic fits span seven orders of magnitude within one rate-law group and depend on each lab's reactor, particle size and activation, so ranking single alloys by them would be misleading. Family activation energies (AB_5 ≈ 31, AB_2 ≈ 24, MgH_2 ≈ 132 kJ/mol) support only a qualitative class.

C6 **Why is the stationary leader TiFe-based and not $LaNi_5$?**

TiFe$_{0.9}$Al$_{0.1}$ has a low, convenient plateau (1.41 bar at 25 °C) and uses cheap elements; LaNi$_5$ has lower capacity (1.49 wt%) and rare-earth cost. With stationary weights (cost 0.25, pressure safety 0.2), TiFe types win.

C7 **What did you actually contribute, if the data are public?**

The audit (finding the sign, unit, leakage and overlap problems), the composition-level dataset, the honestly validated model with a calibrated domain and intervals, the Van't Hoff and PCT consistency analysis, the screening method with sensitivity analysis, and the tested software. Be ready to say which parts you did yourself and which tools you used.

C8 **What is novel here?**

Not the random forest itself. The contribution is the end-to-end, audited workflow: leakage and overlap handled explicitly, distance-calibrated uncertainty attached to every prediction, and a transparent link from ΔH to application decisions with pass/fail/unknown logic and evidence classes.

DEFEND Level 4: traps (don't overclaim)

T1 **"So you discovered new hydrogen storage materials?"**

No. The generated compositions are candidates for measurement. They lie close to known alloys by construction, ΔS is assumed, and capacity is not predicted.

T2 **"Your model predicts that alloy X works at 74 °C."**

The model predicts ΔH only. 74 °C uses an assumed family ΔS, and the band is 15–124 °C. Only a PCT measurement fixes the operating temperature.

T3 **"Your interval means you are 80% sure."**

It is the 10–90% range of past errors for similar distances, an empirical estimate. It is not a guarantee for one alloy, though on a check set 91% of values fell inside.

T4 **"Your best mobile alloy stores 2.1 wt%, so a 1 kg tank needs 48 kg."**

48 kg is the *alloy* mass alone. A real system adds a vessel, a heat exchanger and empty space, so it would be heavier. Also, 2.1 wt% is a reported capacity of unstated definition.

T5 **"You validated on HydPARK, so the model generalises."**

No independent validation set exists; the datasets overlap. Generalisation is estimated by grouped, nested CV within the combined data.

T6 **"Your cost criterion shows alloy A is cheaper."**

It is a qualitative element class (common / rare earth, Co, V / precious), not a price model.

T7 **"You did experiments to confirm this?"**

No experiments were done. All values are literature data; the experimental programme is a proposal.

T8 **"Your PCT check validates the tabulated ΔH."**

It is a consistency check: for 32 of the fits the toolbox authors derived the tabulated value from those same digitised curves.

DEFEND Quick numerical checks they might ask on the board

N1 **What pressure does an alloy with ΔH = 30 kJ/mol, ΔS = 110 J/mol·K give at 25 °C?**

$\ln P = -30000/(8.314 \times 298.15) + 110/8.314 = -12.10 + 13.23 = 1.13 \rightarrow P \approx 3.1$ atm ≈ 3.1 bar.

N2 **Heat to release 1 kg H_2 from a 40 kJ/mol hydride?**

$40/2.016 \approx 19.8$ MJ/kg, about 17% of the LHV.

N3 **Roughly at what temperature does a 40 kJ/mol, 120 J/mol·K hydride give 1 bar?**

$T \approx \Delta H / \Delta S = 40000/120 \approx 333$ K ≈ 60 °C.

N4 **L1 distance between TiFe and $TiFe_{0.9}Mn_{0.1}$?**

Fractions (Ti, Fe, Mn): (0.5, 0.5, 0) vs (0.5, 0.45, 0.05) → 0 + 0.05 + 0.05 = 0.10. Within the domain limit.

Part 10 One-page cheat sheet

Equations

$$\ln \frac{P_{eq}}{P_0} = -\frac{\Delta H}{RT} + \frac{\Delta S}{R}$$

$$T(P) = \frac{\Delta H}{\Delta S - R\ln(P/P_0)}, \quad T(1\,\text{bar}) \approx \frac{\Delta H}{\Delta S}$$

$$Q = \frac{\Delta H}{M_{H_2}}, \;\; 1\ \tfrac{\text{kJ}}{\text{mol}} \approx 0.496\ \tfrac{\text{MJ}}{\text{kg H}_2}$$

$R = 8.314462618$, $P_0 = 1$ atm $= 1.01325$ bar, $M_{H_2} = 2.01588$ g/mol, LHV 119.96 MJ/kg.
Phase rule on plateau: $F = 2 - 3 + 2 = 1$.
5 kJ/mol ≈ ×7.5 in pressure at 25 °C.

Data

- Purdue: 337 .mh files, 68 digitised PCT, negative J/mol
- HydPARK: 772 rows, 4 blocks, positive kJ/mol, ΔS back-calculated (395/412)
- Overlap 248/288 → no external validation
- Union: **725** compositions (276 Purdue, 449 HydPARK); 659 CV groups
- Inter-source scatter ≈ 2.2 kJ/mol

Descriptors

- 38 = 22 elemental stats + 4 alloy + 12 fractions
- Composition only: no ΔS, P, T, capacity
- Top importances: vol_mean 3.57, frac_hydride_former 2.10, vec_std 1.31
- $LaNi_5$ vol_mean = 9.23 cm³/mol

Model

- RF: 300 trees, max_features 0.33, min_samples_leaf 1, seed 0
- Nested grouped CV, 5×3 outer, 4 inner
- **MAE 5.26 ± 0.14**, RMSE 10.32, R^2 0.715
- GB 5.30 · Ridge 7.09 · family median 8.88 · mean 12.26
- Hold-out 5.55 (R^2 0.73, n 145); ungrouped CV 5.12
- AB_5 3.09 · AB_2 4.17; clean subset 3.51
- Leave-family-out: fails (SS 20.98 vs 12.33)

Domain and interval

- MAE 2.62 (d ≤ 0.05) → 16.9 (d > 0.5)
- Domain: $d_{NN} \leq 0.15$
- Interval $[\hat{y} - q_{90},\ \hat{y} - q_{10}]$; 91% coverage on check set
- $LaNi_{4.2}Al_{0.8}$: 38.7 [34.3, 41.6]; $Ti_{0.5}V_{0.5}$: out of domain

Thermodynamics

- $LaNi_5$: 32.15 / 112.8 → 1.84 bar @25 °C, 1 bar @11.6 °C, 15.95 MJ/kg (13.3%)
- TiFe 4.16 bar · TiCrMn 107.5 bar · MgH_2 302 °C, 38.8 MJ/kg
- 85% reach 1 bar ≤ 80 °C; median heat 15.1 MJ/kg
- PCT check median 1.76 kJ/mol (consistency only)
- ML-estimate T band ≈ 105 K (65 ΔH + 35 ΔS)

Screening

- Gate 319/337; mobile pass 232; stationary 252
- Mobile weights: capacity 0.35, heat 0.20 …; stationary: cost 0.25, safety 0.20 …
- Mobile top: Ti-based AB_2/AB; stationary: TiFe-type, Ti–Zr–Mn
- Scenarios: 278/4/55 and 222/6/109; top-5 overlap 0
- Robust membership, not order; τ min 0.65

Never say

- "discovered" → say "candidates"
- "predicts temperature" → "predicts ΔH; ΔS assumed"
- "80% sure" → "empirical interval"
- "tank stores 2 wt%" → "material-level reported capacity"
- "validated externally" → "grouped nested CV"
- "experiments show" → "literature data; experiments proposed"